



Oral Surgery

General Complications of L.A.

Lec. 12

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1) Fainting

- 1.Fainting (Vasovagal Attack)
- It is the **most common systemic complication** that occurs with local anesthesia in the dental office. It refers to a sudden transient loss of consciousness usually secondary to **cerebral ischemia**. The cerebral ischemia is secondary to vasodilatation or an increase in peripheral vascular bed, with a corresponding drop in blood pressure.

1) Fainting

- The collapse in the dental chair may occur suddenly and may not be accompanied by loss of consciousness, in most instances, these episodes are vasovagal attack and spontaneous recovery is usual. The patient often complains of feeling dizzy, weak, and nauseated, the skin is pale, cold and slow pulse is noticed.

Predisposing factors

- The predisposing factors for this condition may be divided into two groups
- 1- Psychogenic factors as
 - A-Fright and anxiety
 - B-Emotional stress
 - C-Pain of sudden and unexpected nature.
 - D-Sight of blood or of surgical or other dental instrument such as local anesthetic syringe and injection needle.

- - Non-psychogenic factors as
- A-Sitting in upright position or standing for prolonged period it leads to pooling of the blood in periphery thereby decreasing cerebral blood flow
- B-Hunger or starvation, which leads to a decrease in cerebral blood glucose level.
- C-Poor physical condition
- D-Hot, humid and crowded environment

Mechanism of Vasovagal Syncope

- Regardless of the trigger, the mechanism of syncope is similar. The brainstem is activated directly or indirectly by the triggering stimulus, resulting in simultaneous enhancement of parasympathetic nervous system (vagal) tone and withdrawal of sympathetic nervous system tone.
- This results in the following responses:

Mechanism of Vasovagal Syncope

- 1) The **cardioinhibitory response**, characterized by a drop-in heart rate and in contractility leading to a decrease in cardiac output that is significant enough to result in a loss of consciousness.
- 2) **Vasodepressor response**: dilation of the blood vessels as a result of the withdrawal of sympathetic nervous system tone. The blood will pool in the dilated peripheral vessels, at the same time, the blood flow to the brain is reduced.

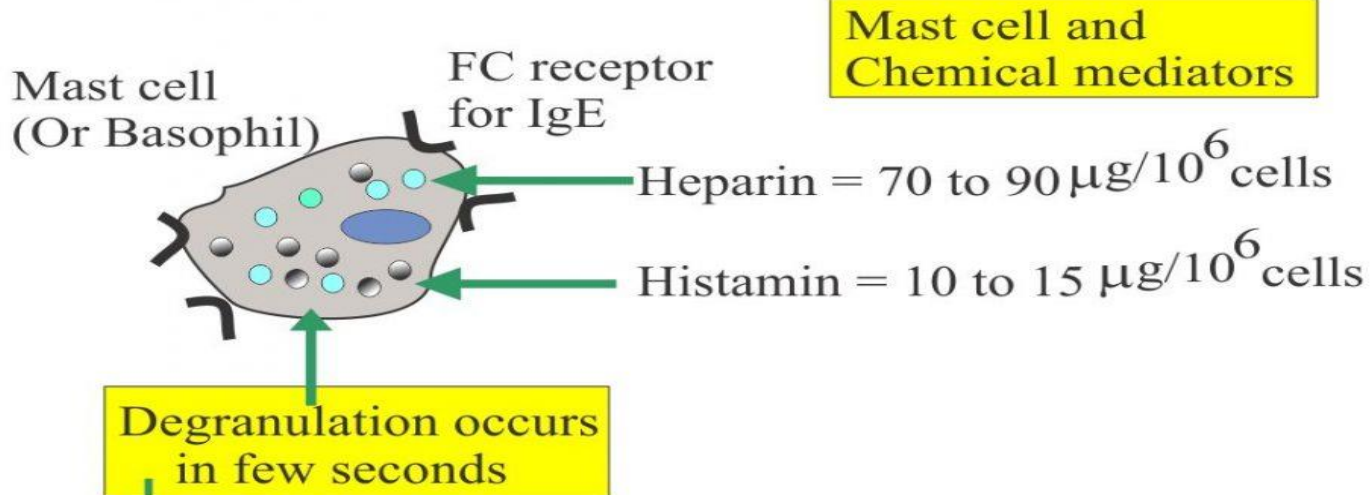
Management

- The head should be **lowered** quickly by adjusting the back of the dental chair so that the patient assume the **supine position with legs elevated**.
- **Tight collar and belts should be loosened** and respiration is stimulated. Spontaneous recovery is usual, but if sign of recovery are not apparent **within 30-60 sec** of the first aid measure, the collapse probably is not a vasovagal attack and in this case medical **emergency team** should be asked immediately for help.

2)Hypersensitivity reaction

- Is defined as an exaggerated immune and inflammatory response to an antigen.
- This phenomenon occurs due to sensitivity of some individual to certain substance known as **allergens**.
- Any local anesthesia agent may evoke such a response, but it is more **commonly seen with ester type agent e.g. Procaine** than amide type e.g. **lidocaine**.
- **True allergy to amide type is extremely rare.**

- Hypersensitivity reaction could be due to:
- 1- Local anesthesia agent.
- 2- Vasoconstrictor.
- 3- Additives: e.g. bisulphate which used as preservative.



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|---------------|---|--|
| Histamine | → | 1. Increase vascular permeability
2. Contraction of smooth muscle |
| Leukotriens | → | Contraction of lung smooth muscles
(Bronchospasm) |
| Serotonine | → | Contraction of smooth muscles |
| ECF-A | → | Eosinophil chemotactic factor |
| Prostagladins | → | 1. Increase vascular permeability
2. Contraction of smooth muscle |
| PAF | → | Release of histamine and serotonin from the platelets |

- In general hypersensitivity reaction to local anesthesia is very rare and represent **less than 1%** of all complications of L.A.
- Allergic response to L.A. may be localized or generalized and immediate or delayed in onset, it may vary from **mild skin irritation or rashes to an anaphylactic reaction.**

- Local reactions are seen more frequently than systemic and usually resolve without active treatment.
- If any degree of allergic reaction is observed, it is very important to determine the actual cause allergen.
- Inadequate diagnoses and treatment can be **life threatening to the patient.**

Prevention

- Proper pre anesthetic evaluation which include proper personal history and the past dental history, particularly history of allergy to local anesthetic agent, or history of allergy to any other drug.

Anaphylactic shock

- Anaphylaxis is a severe, life-threatening allergic reaction. It can happen seconds or minutes after exposure to allergen.
- It causes hypotension, air way obstruction in association with laryngeal edema unless treatment is initiated immediately; the condition may progress to fatal termination.

Clinical presentation

- It is characterized by:
- 1- Profound fall in blood pressure.
- 2 -Dyspnea and respiratory embarrassment
- 3- Facial and laryngeal edema
- 4-Loss of consciousness.
- 5- Urticaria.
- 6- Nausea, vomiting, and diarrhea.



Management

- -If suspicion of the anaphylactic reaction occurring, immediately seek medical help. The treatment must begin before blood pressure and breathing problem become life threatening.
- -**Epinephrine is the most important medication** for the treatment of anaphylaxis; it is injection into a muscle. Epinephrine works rapidly to make blood vessel contract, preventing them from leaking more fluid. It also relax airways helping the individual breathe easier, relieves cramping in the GIT and stop itching, rapidly reverse the life threatening symptom of anaphylactic shock, if given in time.



- Even if the individual responds to the epinephrine, it is vitally important to go to an emergency room immediately.
- O2 and additional medication as antihistamine to counteract the effect of histamine and help in prevention of delayed allergic reaction.
- Corticosteroids are another treatment option especially the rapid acting like hydrocortisone.
- -IV fluid may be necessary to restore adequate blood pressure.

3)Over dosage and toxicity

- It is relatively rare, a toxicity reaction can occur when the concentration of local anesthetic solution in circulation increase too rapidly within a short period of time e.g. injection too rapidly into highly vascular area or IV injection.
- The toxicity varies among patients and is influenced by numerous factors which are:
 - 1-Amount and type of local anesthesia administered.
 - 2- The patient general health, age and weight
 - 3 -Rout of administration
 - 4- Rapidity of injection.

- Mild over dosage sign and symptom: restlessness, retention of consciousness, talkativeness, and agitation, which may end into convulsion along with increased BP, heart rate and respiratory rate.

Management

- 1-Stop the dental procedure
- 2-Position the patient supine with legs elevated
- 3-Reassurance to the patient
- 4-administer O₂, IV anticonvulsant as diazepam
- 5- if the patient recovers then discharge, but when the patient fails to recover transfer him to a general hospital

4)Drug interaction

- In some patients the administration of 2 drugs will **counteract** each other while in other **potentiation occurs**. In patient using **tricyclic antidepressant** (ex: Imipramine) variable degrees of **potentiation of BP response to adrenaline and nor adrenaline will occur**.
- With patient using tricyclic antidepressant the practitioners can minimize the risk of interaction by using an **aspirating syringe**, which reduces the likelihood of the local anesthetic being administered directly into a blood vessel or **using L.A solution containing other vasoconstrictor as felypressin**.

- An additional concern is the potential for adverse drug interactions between vasoconstrictors and the nonselective β -adrenergic blocking agents (e.g., propranolol) that may result in uncompensated peripheral vasoconstriction.